11 (a) Solve $3x^2 = 108$

$3x^2 = 108$

$\div 3 \qquad \div 3$

$x^2 = 36$, so $x = -6$ or 6

1 mark
Isolate x^2.

1 mark
Both roots.

(2)

(b) Factorise $x^2 - 2x - 35$

	x	5
x	x^2	$5x$
-7	$-7x$	-35

$-7 \times 5 = -35$; $-7 + 5 = -2$

$(x - 7)(x + 5)$

1 mark
Find factor pair.

1 mark
Correct signs.

(2)

$u = \dfrac{3t}{4} + 2$

(c) Make t the subject of the formula.

$u - 2 = \dfrac{3t}{4}$

$4(u - 2) = 3t$

$t = \dfrac{4(u - 2)}{3}$

1 mark
Subtract 2 from both sides.

1 mark
Multiply both sides by 4.

1 mark
Divide by 3.

(3)

(Total for Question 11 is 7 marks)